

Problem definition

Design the database for StudentHub with students, events and registrations. Create PHP database connection using MySQL statements.

Key Questions

1. Is the schema modeled properly?

Yes, the StudentHub database schema is modeled properly. It contains three main tables: students, events, and registrations. The registrations table connects students and events using foreign keys and represents the many-to-many relationship between students and events.

2. Are primary keys and foreign keys correctly defined?

Yes. Each table has a properly defined primary key. `student_id`, `event_id`, and `registration_id` are primary keys in their respective tables. The `student_id` and `event_id` in the registrations table are foreign keys referencing the students and events tables.

3. Is the database connection handled securely?

Yes. The PHP application uses PDO for database connectivity and Prepared Statements for executing SQL queries. Prepared Statements help prevent SQL Injection attacks and provide secure database operations. Error handling is also implemented using try-catch.

Code:

Db.php

```
<?php

$host = "localhost";
$dbname = "studenthub";
$username = "root";
$password = "";

try {

    $pdo = new PDO(
        "mysql:host=$host;dbname=$dbname",
        $username,
        $password
    );
```

```
$pdo->setAttribute(
    PDO::ATTR_ERRMODE,
    PDO::ERRMODE_EXCEPTION
);

echo "Database Connected Successfully";

}
catch (PDOException $e) {

    echo "Connection Failed: " . $e->getMessage();

}

?>
```

Add_student.php

```
<?php

include "db.php";

$name = "Jay";
$email = "jay@gmail.com";
$phone = "9876543210";

$sql = "INSERT INTO students (name, email, phone)
        VALUES (?, ?, ?)";

$stmt = $pdo->prepare($sql);

$stmt->execute([
    $name,
    $email,
    $phone
]);

echo "Student Added Successfully";

?>
```

Add_event.php

```
<?php

include "db.php";

$event_name = "AI Workshop";
$event_date = "2026-12-01";
$venue = "Seminar Hall";

$sql = "INSERT INTO events (event_name, event_date, venue)
        VALUES (?, ?, ?)";

$stmt = $pdo->prepare($sql);

$stmt->execute([
    $event_name,
    $event_date,
    $venue
]);

echo "Event Added Successfully";

?>
```

Register.php

```
<?php

include "db.php";

$student_id = 4;
$event_id = 4;

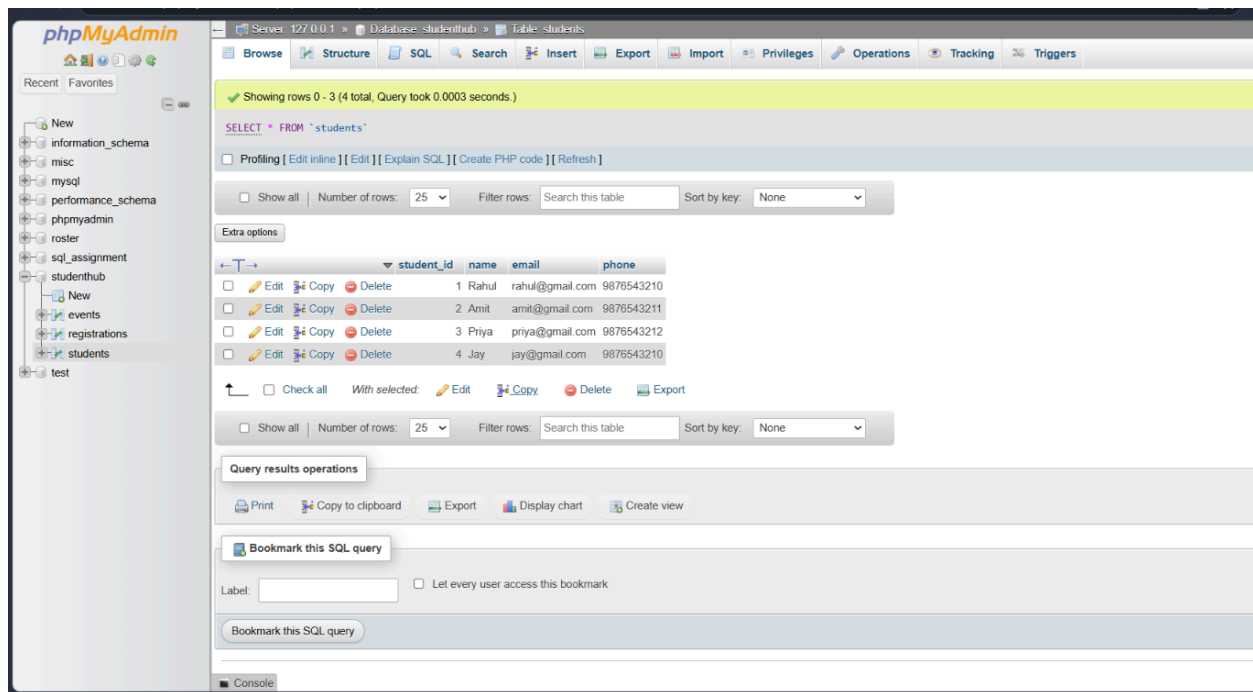
$sql = "INSERT INTO registrations (student_id, event_id)
        VALUES (?, ?)";

$stmt = $pdo->prepare($sql);

$stmt->execute([
    $student_id,
    $event_id
]);
```

```
] );  
  
echo "Registration Successful";  
  
?>
```

Output



The screenshot shows the phpMyAdmin interface for a database named 'studenthub'. The 'students' table is selected, and the SQL query 'SELECT * FROM `students`' is executed. The results show 4 rows of student data. The interface includes a sidebar with a database tree, a top navigation bar, and a bottom console.

Showing rows 0 - 3 (4 total. Query took 0.0003 seconds.)

SELECT * FROM `students`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	student_id	name	email	phone
☐	1	Rahul	rahul@gmail.com	9876543210
☐	2	Amit	amit@gmail.com	9876543211
☐	3	Priya	priya@gmail.com	9876543212
☐	4	Jay	jay@gmail.com	9876543210

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

The screenshot shows the phpMyAdmin interface for the 'studenthub' database. The 'events' table is selected, and the SQL query 'SELECT * FROM `events`' is displayed. The table contains 4 rows of data.

event_id	event_name	event_date	venue
1	Hackathon	2026-10-20	Seminar Hall
2	Tech Fest	2026-11-05	Auditorium
3	Coding Contest	2026-11-15	Lab 1
4	AI Workshop	2026-12-01	Seminar Hall

The screenshot shows the phpMyAdmin interface for the 'studenthub' database. The 'registrations' table is selected, and the SQL query 'SELECT * FROM `registrations`' is displayed. The table contains 5 rows of data.

registration_id	student_id	event_id	registration_date
1	1	1	2026-10-03 23:43:29
2	1	2	2026-10-03 23:43:29
3	2	1	2026-10-03 23:43:29
4	3	3	2026-10-03 23:43:29
5	4	4	2026-10-03 23:55:56

localhost/studenthub/practical/view_registrations.php

Database Connected Successfully Student: Rahul Email: rahul@gmail.com Event: Hackathon Event Date: 2026-10-20 Venue: Seminar Hall Registration Date: 2026-10-03 23:43:29
Student: Rahul Email: rahul@gmail.com Event: Tech Fest Event Date: 2026-11-05 Venue: Auditorium Registration Date: 2026-10-03 23:43:29
Student: Amit Email: amit@gmail.com Event: Hackathon Event Date: 2026-10-20 Venue: Seminar Hall Registration Date: 2026-10-03 23:43:29
Student: Priya Email: priya@gmail.com Event: Coding Contest Event Date: 2026-11-15 Venue: Lab 1 Registration Date: 2026-10-03 23:43:29
Student: Jay Email: jay@gmail.com Event: AI Workshop Event Date: 2026-12-01 Venue: Seminar Hall Registration Date: 2026-10-03 23:55:56